MODULE *IntegrationContract*

Section: Imports

Shared operators used by this module are imported below.

EXTENDS *Naturals*, *FiniteSets*, *FailurePolicy*

Section: Integration boundary

This layer receives completed per-check snapshots from *Overwatch*. It specifies requests, responses, delivery channels, and session lifecycle for every host. Both layers use the same *FailurePolicy* definitions.

CONSTANTS *Hosts*, *Checks*, *Names*, *MaxRequests*, *MaxChanges*, *MaxSessions*

$AllHosts \triangleq \{\text{"claude"}, \text{"codex"}, \text{"opencode1"}, \text{"opencode2"}, \text{"pi"}, \text{"neovim"}\}$
ASSUME $Hosts \subseteq AllHosts \land Hosts \neq \{\}$

These are host capabilities, not advisory/strict policy modes. A warning cannot be counted as a denied completion, and an editor has no agent gate.

Section: Host capabilities

The host determines whether protected requests wait for a running check, how guarded completion is handled, and which delivery channels exist. These are integration capabilities, not user-selectable enforcement modes.

$$
\begin{aligned}
&Capabilities(h) \triangleq \\
&\quad [agent \mapsto h \neq \text{"neovim"}, \\
&\quad\ waitsForCheck \mapsto h \in \{\text{"codex"}, \text{"opencode1"}, \text{"opencode2"}, \text{"pi"}\}, \\
&\quad\ completion \mapsto \text{CASE } h \in \{\text{"claude"}, \text{"codex"}\} \to \text{"block"} \\
&\qquad\qquad\qquad \square\ \ h = \text{"opencode1"} \to \text{"warn"} \\
&\qquad\qquad\qquad \square\ \ h = \text{"opencode2"} \to \text{"queue"} \\
&\qquad\qquad\qquad \square\ \ h = \text{"pi"} \to \text{"continue"} \\
&\qquad\qquad\qquad \square\ \ \text{OTHER} \to \text{"none"}, \\
&\quad\ delivery \mapsto \text{CASE } h = \text{"claude"} \to \text{"turn"} \\
&\qquad\qquad\qquad \square\ \ h = \text{"opencode2"} \to \text{"context"} \\
&\qquad\qquad\qquad \square\ \ h = \text{"neovim"} \to \text{"quickfix"} \\
&\qquad\qquad\qquad \square\ \ \text{OTHER} \to \text{"event"}, \\
&\quad\ idleWake \mapsto h \in \{\text{"claude"}, \text{"opencode2"}\}, \\
&\quad\ status \mapsto h \in \{\text{"claude"}, \text{"pi"}, \text{"neovim"}\}]
\end{aligned}
$$

$$
\begin{aligned}
&Kinds \triangleq \{\text{"inspect"}, \text{"repair"}, \text{"publish"}, \text{"complete"}\} \\
&Empty \triangleq [c \in Checks \mapsto \{\}] \\
&Snapshots \triangleq [Checks \to \text{SUBSET } Names] \\
&NoReply \triangleq [ready \mapsto \text{FALSE}, decision \mapsto \text{"none"}]
\end{aligned}
$$

Section: State shape

Request and reply are separate so that a check can finish or a session can close between them. Report visibility, deferred context, corrections, and idle wakeups are separate observations.

VARIABLES *host*, *active*, *busy*, *reports*, *baseline*, *inherited*, *selected*,
request, *reply*, *requests*, *changes*, *session*,
visible, *contextPending*, *contextSent*, *correctionQueued*,
contexts, *warnings*, *corrections*, *inTurn*, *wakes*, *work*

$$
\begin{aligned}
vars \triangleq \langle &host, active, busy, reports, baseline, inherited, selected, \\
&request, reply, requests, changes, session, visible, \\
&contextPending, contextSent, correctionQueued, contexts, \\
&warnings, corrections, inTurn, wakes, work\rangle
\end{aligned}
$$

Section: *Guard* selection and inherited policy

An empty selection means all checks. *FailureGuards* decides whether each selected check guards publication; *Notify* identifies new failures for delivery.

$$
\begin{aligned}
&Guarded(scope) \triangleq \\
&\quad \{c \in scope : FailureGuards(reports[c], baseline[c], inherited)\}
\end{aligned}
$$

```
Guard ≜ Capabilities(host).agent ∧ Guarded(Checks) ≠ {}
Notify ≜ Capabilities(host).agent ∧
         (∃ c ∈ Checks : FailureGuards(reports[c], baseline[c], "allow"))
SelectedChecks ≜ IF selected = {} THEN Checks ELSE selected
```

Section: State domains

Counters and snapshots are bounded for *TLC* exploration. These bounds do not assert a runtime limit on requests or sessions.

```
TypeOK ≜
    ∧ host ∈ Hosts ∧ active ∈ BOOLEAN ∧ busy ∈ BOOLEAN
    ∧ work ∈ 0 .. ((MaxSessions + 1) * MaxRequests)
    ∧ inTurn ∈ BOOLEAN
    ∧ wakes ∈ 0 .. (MaxChanges + MaxSessions + 1 + (MaxSessions + 1) * MaxRequests)
    ∧ reports ∈ Snapshots ∧ baseline ∈ Snapshots
    ∧ inherited ∈ {"allow", "block"} ∧ selected ⊆ Checks
    ∧ request ∈ Kinds ∪ {"none"}
    ∧ reply ∈ [ready : BOOLEAN ,
               decision : {"none", "allow", "deny", "block", "warn", "queue", "continue", "cancelled"}]
    ∧ requests ∈ 0 .. MaxRequests ∧ changes ∈ 0 .. MaxChanges ∧ session ∈ 0 .. MaxSessions
    ∧ visible ∈ [known : BOOLEAN , failures : Snapshots]
    ∧ contextPending ∈ BOOLEAN ∧ contextSent ∈ BOOLEAN ∧ correctionQueued ∈ BOOLEAN
    ∧ contexts ∈ 0 .. (MaxChanges + MaxSessions + 1)
    ∧ warnings ∈ 0 .. ((MaxSessions + 1) * MaxRequests)
    ∧ corrections ∈ 0 .. ((MaxSessions + 1) * MaxRequests)
```

Section: Initial state

A host starts active with arbitrary reports, baseline, policy, and check state. No reply, visible report, deferred context, or correction has been produced.

```
Init ≜
    ∧ host ∈ Hosts ∧ active = TRUE ∧ busy ∈ BOOLEAN
    ∧ reports ∈ Snapshots ∧ baseline ∈ Snapshots
    ∧ inherited ∈ {"allow", "block"} ∧ selected ∈ SUBSET Checks
    ∧ request = "none" ∧ reply = NoReply
    ∧ requests = 0 ∧ changes = 0 ∧ session = 0
    ∧ visible = [known ↦ FALSE, failures ↦ Empty]
    ∧ contextPending = FALSE ∧ contextSent = FALSE ∧ correctionQueued = FALSE
    ∧ contexts = 0 ∧ warnings = 0 ∧ corrections = 0 ∧ wakes = 0 ∧ work = 0 ∧ inTurn ∈ BOOLEAN
```

Section: Requests and completion decisions

Inspection and repair remain answerable during checks. Protected requests follow the host waiting capability and consult the selected reports. Respond records the decision and accounts separately for warnings, corrections, and wakeups; *Acknowledge* releases the request slot.

```
Request(k) ≜
    ∧ active ∧ request = "none" ∧ requests < MaxRequests
    ∧ request' = k ∧ reply' = NoReply ∧ requests' = requests + 1
    ∧ UNCHANGED ⟨host, active, busy, reports, baseline, inherited, selected,
                 changes, session, visible, contextPending, contextSent,
                 correctionQueued, contexts, warnings, corrections, inTurn, wakes, work⟩

CanRespond ≜
    active ∧ request ≠ "none" ∧ ¬reply.ready ∧
    (request ∈ {"inspect", "repair"}      ∨ ¬Capabilities(host).agent ∨
     ¬Capabilities(host).waitsForCheck ∨ ¬busy)

Decision ≜
    CASE request ∈ {"inspect", "repair"} → "allow"
      □ request = "publish" →
          IF Capabilities(host).agent ∧ Guarded(SelectedChecks) ≠ {}
```

```
                THEN "deny" ELSE "allow"
        □   request = "complete" → IF Guard THEN Capabilities(host).completion ELSE "allow"

QueuesCorrection ≜ Decision = "queue" ∨ (host = "claude" ∧ Decision = "block")

Respond ≜
    ∧ CanRespond
    ∧ inTurn' = IF request = "complete" ∧
                    (Decision ∈ {"allow", "warn", "queue"} ∨ host = "claude")
                THEN FALSE ELSE inTurn
    ∧ reply' = [ready ↦ TRUE, decision ↦ Decision]
    ∧ warnings' = warnings + IF Decision = "warn" THEN 1 ELSE 0
    ∧ corrections' = corrections + IF QueuesCorrection ∧ ¬correctionQueued THEN 1 ELSE 0
    ∧ correctionQueued' = (correctionQueued ∨ QueuesCorrection)
    ∧ wakes' = wakes + IF QueuesCorrection ∧ ¬correctionQueued ∧
                          (host = "opencode2" ∨ ¬inTurn)
                       THEN 1 ELSE 0
    ∧ UNCHANGED ⟨host, active, busy, reports, baseline, inherited, selected,
                 request, requests, changes, session, visible,
                 contextPending, contextSent, contexts, work⟩

Acknowledge ≜
    ∧ request ≠ "none" ∧ reply.ready
    ∧ request' = "none"
    ∧ UNCHANGED ⟨host, active, busy, reports, baseline, inherited, selected,
                 reply, requests, changes, session, visible,
                 contextPending, contextSent, correctionQueued,
                 contexts, warnings, corrections, inTurn, wakes, work⟩
```

Section: *Checks* and result delivery

A completed check replaces the reports. Changed findings invalidate old delivery and correction deduplication. Report updates visibility and either publishes event context or retains it for *Deliver*. Recovery clears stale context.

```
BeginCheck ≜
    ∧ active ∧ ¬busy ∧ changes < MaxChanges
    ∧ busy' = TRUE
    ∧ UNCHANGED ⟨host, active, reports, baseline, inherited, selected,
                 request, reply, requests, changes, session, visible,
                 contextPending, contextSent, correctionQueued,
                 contexts, warnings, corrections, inTurn, wakes, work⟩
```

The result-ordering and baseline rules are specified in *Overwatch.tla*. This layer accepts a completed result at that boundary, then constrains what every host must expose and which requests it may hold behind a check.

```
FinishCheck(f) ≜
    ∧ active ∧ busy ∧ changes < MaxChanges
    ∧ reports' = f ∧ busy' = FALSE ∧ changes' = changes + 1
    ∧ IF f ≠ reports
      THEN ∧ contextSent' = FALSE ∧ correctionQueued' = FALSE
           ∧ contextPending' = FALSE
      ELSE UNCHANGED ⟨contextSent, correctionQueued, contextPending⟩
    ∧ UNCHANGED ⟨host, active, baseline, inherited, selected, request,
                 reply, requests, session, visible, contexts, warnings, corrections, inTurn, wakes, work⟩

Report ≜
    ∧ active ∧ ¬busy
    ∧ visible' = [known ↦ TRUE, failures ↦ reports]
    ∧ contextPending' = (Notify ∧ ¬contextSent ∧ Capabilities(host).delivery ∈ {"turn", "context"})
    ∧ contextSent' = (contextSent ∨ (Notify ∧ Capabilities(host).delivery = "event"))
    ∧ contexts' = contexts + IF Notify ∧ ¬contextSent ∧ Capabilities(host).delivery = "event" THEN 1 ELSE 0
```

$\land$ UNCHANGED $\langle host, active, busy, reports, baseline, inherited, selected,$
$request, reply, requests, changes, session,$
$correctionQueued, warnings, corrections, inTurn, wakes, work\rangle$

$Deliver \triangleq$
$\land\ active \land contextPending$
$\land\ wakes' = wakes +$ IF $host =$ "claude" $\land \neg inTurn$ THEN 1 ELSE 0
$\land\ contextSent' =$ TRUE $\land\ contextPending' =$ FALSE $\land\ contexts' = contexts + 1$
$\land$ UNCHANGED $\langle host, active, busy, reports, baseline, inherited, selected,$
$request, reply, requests, changes, session, visible,$
$correctionQueued, warnings, corrections, inTurn, work\rangle$

Section: Shutdown restart and new work

Stop cancels an outstanding request and clears visible and pending delivery. Restart preserves project policy, takes reports as the new baseline, and resets delivery state. Completed tool work permits a fresh correction.

$Stop \triangleq$
$\land\ active$
$\land\ inTurn' =$ FALSE
$\land\ active' =$ FALSE $\land\ busy' =$ FALSE $\land\ request' =$ "none"
$\land\ reply' = [ready \mapsto$ TRUE, $decision \mapsto$ "cancelled"$]$
$\land\ visible' = [known \mapsto$ FALSE, $failures \mapsto Empty]$
$\land\ contextPending' =$ FALSE
$\land$ UNCHANGED $\langle host, reports, baseline, inherited, selected, requests,$
$changes, session, contextSent, correctionQueued,$
$contexts, warnings, corrections, wakes, work\rangle$

$Restart \triangleq$
$\land \neg active \land session < MaxSessions$
$\land\ inTurn' =$ FALSE
$\land\ active' =$ TRUE $\land\ session' = session + 1 \land\ busy' =$ FALSE
$\land\ baseline' = reports$
$\land\ request' =$ "none" $\land\ reply' = NoReply \land\ requests' = 0$
$\land\ visible' = [known \mapsto$ FALSE, $failures \mapsto Empty]$
$\land\ contextPending' =$ FALSE $\land\ contextSent' =$ FALSE $\land\ correctionQueued' =$ FALSE
$\land$ UNCHANGED $\langle host, reports, inherited, selected, changes, contexts, warnings, corrections, wakes, work\rangle$

$StartTurn \triangleq$
$\land\ active \land \neg inTurn \land inTurn' =$ TRUE
$\land$ UNCHANGED $\langle host, active, busy, reports, baseline, inherited, selected,$
$request, reply, requests, changes, session, visible,$
$contextPending, contextSent, correctionQueued, contexts,$
$warnings, corrections, wakes, work\rangle$

$WorkCompleted \triangleq$
$\land\ active \land work < (MaxSessions + 1) * MaxRequests$
$\land\ work' = work + 1 \land\ correctionQueued' =$ FALSE
$\land$ UNCHANGED $\langle host, active, busy, reports, baseline, inherited, selected,$
$request, reply, requests, changes, session, visible,$
$contextPending, contextSent, contexts, warnings, corrections, inTurn, wakes\rangle$

Section: Allowed executions and fairness

Spec composes check-completion fairness with fair response and report servicing. Inspection independence is a safety property: its response remains enabled even while a check is held.

$CheckCompletes \triangleq \exists f \in Snapshots : FinishCheck(f)$
$Next \triangleq (\exists k \in Kinds : Request(k)) \lor Respond \lor Acknowledge \lor BeginCheck$
$\lor CheckCompletes \lor Report \lor Deliver \lor Stop \lor Restart \lor StartTurn \lor WorkCompleted$
$\lor (\neg active \land session = MaxSessions \land$ UNCHANGED $vars)$

Check completion fairness is the *RunsEnd* guarantee of the result model. Inspection is independently enabled; it never needs that guarantee to reply.

$$Spec \triangleq Init \land \Box[Next]_{vars} \land \mathrm{WF}_{vars}(CheckCompletes) \land \mathrm{WF}_{vars}(Respond) \land \mathrm{WF}_{vars}(Report) \land \mathrm{WF}_{vars}(Deliver)$$

Section: Safety and progress properties

Capabilities constrain completion, editors never acquire agent gates, closed sessions stay silent, pending context stays current, and corrections are deduplicated. Progress properties require the fairness stated in *Spec*.

$$InspectionIndependentOfChecks \triangleq active \land request \in \{\text{“inspect”}, \text{“repair”}\} \land \neg reply.ready \Rightarrow \text{ENABLED } Respond$$

$$InspectionAlwaysAllowed \triangleq request \in \{\text{“inspect”}, \text{“repair”}\} \land reply.ready \Rightarrow reply.decision = \text{“allow”}$$

$$EditorNeverGuards \triangleq host = \text{“neovim”} \land request \neq \text{“none”} \land reply.ready \Rightarrow reply.decision = \text{“allow”}$$

$$StoppedSessionsStaySilent \triangleq \neg active \Rightarrow \neg contextPending \land \neg visible.known$$

$$HostsWithoutIdleWakeStayIdle \triangleq \neg Capabilities(host).idleWake \Rightarrow wakes = 0$$

$$CorrectionRequestsAreDeduplicated \triangleq corrections \leq changes + session + work + 1$$

$$PendingContextIsCurrent \triangleq contextPending \Rightarrow Notify$$

$$CompletionUsesHostCapability \triangleq request = \text{“complete”} \land reply.ready \Rightarrow reply.decision \in \{\text{“allow”}, Capabilities(host).completion\}$$

$$EditorOnlyReports \triangleq host = \text{“neovim”} \Rightarrow contexts = 0 \land warnings = 0 \land corrections = 0 \land wakes = 0$$

$$PendingReportsEventuallyResolve \triangleq contextPending \leadsto \neg contextPending$$

$$FinishedReportsBecomeVisible \triangleq (active \land \neg busy) \leadsto (visible.known \lor \neg active)$$

$$InspectionEventuallyReplies \triangleq (active \land request \in \{\text{“inspect”}, \text{“repair”}\} \land \neg reply.ready) \leadsto (reply.ready \lor \neg active)$$